

(1) $E_n = -\frac{1}{4\pi\epsilon_0} \frac{2\alpha_0}{n^2} \frac{e^2}{n^2}$ $\alpha_0 = \frac{4\pi\epsilon_0\hbar^2}{m_0 e^2}$ n^2
 (2) 自旋轨道耦合; 核磁矩足 离子实

(3) $E_n = -\frac{Z^2}{4\pi\epsilon_0\epsilon_r} \frac{e^2}{2a} \frac{1}{n^2} = -\frac{Z^2}{\epsilon_r a_0} \frac{e^2}{2a_0} \frac{1}{n^2} \frac{1}{4\pi\epsilon_0} = -\frac{1}{3 \times 0.07} \frac{-13.6\text{eV}}{n^2} = -\frac{6.5\text{eV}}{n^2}$

(=) (1) $\psi(a+0) = \psi(a+0^-)$, $\psi'(a+0^+) - \psi'(a+0^-) = \frac{2\mu A}{\hbar^2} \psi(a)$

(2) $B e^{-K_1 a} = \cosh K_1 a$, $-K_1 B e^{-K_1 a} - K_1 \sinh K_1 a = \frac{2\mu A}{\hbar^2} B e^{-K_1 a}$
 $-C e^{-K_2 a} = \sinh K_2 a$, $K_2 C e^{-K_2 a} - K_2 \cosh K_2 a = \frac{2\mu A}{\hbar^2} - C e^{-K_2 a}$

(3)

(4) $K_1 < K_2$ 有解

(=) $S_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} \hbar$ $S_x = \frac{\hbar}{\sqrt{2}} \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix}$

$\hat{H} = A \hbar \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix} + \frac{B \hbar^2}{2} \begin{pmatrix} 1 & 0 & 1 \\ 0 & 2 & 0 \\ 1 & 0 & 1 \end{pmatrix}$

$= \begin{pmatrix} A\hbar + \frac{B}{2}\hbar^2 & 0 & \frac{B\hbar^2}{2} \\ 0 & B\hbar^2 & 0 \\ \frac{B\hbar^2}{2} & 0 & -A\hbar + \frac{B}{2}\hbar^2 \end{pmatrix}$

$\begin{vmatrix} A\hbar + \frac{B}{2}\hbar^2 - E_n & 0 & \frac{B\hbar^2}{2} \\ 0 & B\hbar^2 - E_n & 0 \\ \frac{B\hbar^2}{2} & 0 & -A\hbar + \frac{B}{2}\hbar^2 - E_n \end{vmatrix} = 0$

$(B\hbar^2 - E_n) \left[(A\hbar + \frac{B}{2}\hbar^2 - E_n)(-A\hbar + \frac{B}{2}\hbar^2 - E_n) - \frac{B\hbar^2}{2} \frac{B\hbar^2}{2} \right] = 0$

$(B\hbar^2 - E_n) \left[(\frac{B}{2}\hbar^2 - E_n)^2 - (A\hbar)^2 - \frac{B\hbar^2}{2} \frac{B\hbar^2}{2} \right] = 0$

$E_1 = B\hbar^2$

$E_2 = \frac{B\hbar^2}{2} + \sqrt{(A\hbar)^2 + \frac{B^2\hbar^4}{4}}$

$E_3 = \frac{B\hbar^2}{2} - \sqrt{(A\hbar)^2 + \frac{B^2\hbar^4}{4}}$



$$\frac{1}{2} A \hbar = a \quad \frac{1}{2} = b \quad C = \sqrt{a^2 + b^2}$$

$$E_1 = 2b$$

$$E_2 = b + C$$

$$E_3 = b - C$$

$$\begin{pmatrix} a+b & 0 & b \\ 0 & 2b & 0 \\ b & 0 & -a+b \end{pmatrix}$$

$$\psi_1 = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}$$

$$\psi_2 = \begin{pmatrix} a \\ 0 \\ b \end{pmatrix} \frac{1}{\sqrt{b^2 + (a+C)^2}}$$

$$\psi_3 = \begin{pmatrix} a-C \\ 0 \\ b \end{pmatrix} \frac{1}{\sqrt{b^2 + (a-C)^2}}$$

$$|\psi(0)\rangle = C_1 \psi_1 + C_2 \psi_2 + C_3 \psi_3$$

$$|\psi(t)\rangle = C_1 \psi_1 e^{-i\frac{E_1}{\hbar}t} + C_2 \psi_2 e^{-i\frac{E_2}{\hbar}t} + C_3 \psi_3 e^{-i\frac{E_3}{\hbar}t}$$

$$C_1 = \langle \psi_1 | \psi(0) \rangle = 0$$

$$C_2 = \langle \psi_2 | \psi(0) \rangle = \frac{a+C}{\sqrt{b^2 + (a+C)^2}}$$

$$C_3 = \langle \psi_3 | \psi(0) \rangle = \frac{a-C}{\sqrt{b^2 + (a-C)^2}}$$

$$P = |\langle \psi | \psi(t) \rangle|^2 = \left| \frac{(a+C)b}{b^2 + (a+C)^2} e^{-i\frac{E_2}{\hbar}t} + \frac{b(a-C)}{b^2 + (a-C)^2} e^{-i\frac{E_3}{\hbar}t} \right|^2$$

$$= \frac{a-C}{b} + \frac{b^2}{2C^2} \cos \frac{2Ct}{\hbar}$$

$$(1) \int_{-\infty}^{+\infty} |A|^2 e^{-\alpha^2 x^2} dx = 1$$

$$|A|^2 = \frac{2}{\sqrt{\pi}} \quad A = \sqrt{\frac{2}{\sqrt{\pi}}}$$

$$(2) \int_{-\infty}^{+\infty} \frac{2}{\sqrt{\pi}} e^{-\frac{\alpha^2 x^2}{2}} \chi \sqrt{\frac{2}{\sqrt{\pi}}} e^{-\frac{\alpha^2 x^2}{2}} dx = 0$$

$$\langle \hat{p} \rangle = \langle \psi | \hat{p} | \psi \rangle = \int_{-\infty}^{+\infty} \sqrt{\frac{2}{\sqrt{\pi}}} e^{-\frac{\alpha^2 x^2}{2}} \frac{1}{i\hbar} \left(\frac{d}{dx} e^{-\frac{\alpha^2 x^2}{2}} \right) dx = 0$$

$$\langle x^2 \rangle = \int_{-\infty}^{+\infty} x^2 \frac{2}{\sqrt{\pi}} e^{-\frac{\alpha^2 x^2}{2}} dx = \int_0^{\infty} \frac{2\alpha}{\sqrt{\pi}} x^2 e^{-\frac{\alpha^2 x^2}{2}} dx$$

$$\text{令 } t = \frac{\alpha^2 x^2}{2}$$

$$\langle \hat{p}^2 \rangle = \int_{-\infty}^{+\infty} \hbar^2 \left| \frac{d}{dx} \left(e^{-\frac{\alpha^2 x^2}{2}} \right) \right|^2 \frac{2}{\sqrt{\pi}} dx = \frac{1}{\sqrt{\pi}} \alpha^2 \int_0^{\infty} e^{-t} t^2 dt = \frac{1}{\sqrt{\pi}} \alpha^2 \frac{\Gamma(\frac{3}{2})}{2} = \frac{1}{2\alpha^2}$$

$$= \frac{\hbar^2 \alpha^2}{2} \int_0^{\infty} e^{-t} t^2 dt = \frac{\hbar^2 \alpha^2}{2} \frac{\Gamma(\frac{3}{2})}{2\alpha^3} = \frac{\hbar^2 \alpha^2}{2}$$

$$(3) \psi(x) = \sqrt{\frac{2}{\sqrt{\pi}}} e^{-\frac{1}{2} \alpha^2 x^2}$$

$$(4) \langle x^2 \rangle \langle \hat{p}^2 \rangle = \frac{\hbar^2}{4} \geq \frac{\hbar^2}{4}$$



$$\text{五. } \begin{cases} a^\dagger = \frac{1}{\sqrt{2m\hbar\omega}} (\hat{p} + im\omega\hat{x}) \\ a = \frac{1}{\sqrt{2m\hbar\omega}} (\hat{p} - im\omega\hat{x}) \end{cases} \Rightarrow \begin{cases} \hat{p} = i\frac{m\hbar\omega}{2} (a^\dagger - a) \\ \hat{x} = -i\sqrt{\frac{\hbar}{2m\omega}} (a^\dagger + a) \end{cases}$$

$$\text{六. } \begin{aligned} E'_1 &= E_1 + 2 + \frac{|B_2|^2}{E_1 - E_2} & |\psi'_1\rangle &= |\psi_1\rangle + \frac{B_2}{E_1 - E_2} |\psi_2\rangle \\ E'_2 &= E_2 + 2 + \frac{|B_1|^2}{E_2 - E_1} & |\psi'_2\rangle &= |\psi_2\rangle + \frac{B_1}{E_2 - E_1} |\psi_1\rangle \end{aligned}$$

$$H' = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$$

$$\phi = C_1 |\psi_1\rangle + C_2 |\psi_2\rangle$$

$$\begin{vmatrix} 2 - E_n^{(1)} & 0 \\ 0 & 2 - E_n^{(1)} \end{vmatrix} = 0 \quad E_n^{(1)} = 2$$

$$E = E_1 + 2 \leftarrow \text{修正能量}$$

$$\phi = C_1 |\psi_1\rangle + C_2 |\psi_2\rangle \leftarrow \text{零级波}$$

